Section-3

SURGE ARRESTORS

1.0 SCOPE: Refer Clause-1 of Section-I, General

2.0 STANDARDS:

2.1 The Surge Arrestors shall conform to the latest revisions with amendments available of relevant standards, rules and codes some of which are listed herein for ready reference, unless otherwise specified elsewhere in this specification.

Sl.No.	Title	Standard Ref. No
a)	Application Guide for Metal Oxide : Surge Arrester for AC System 2013	IEEE Std.C62-22
b)	Lightning Arresters :	IS:15086 Part-4:2017 IEC-60099-4&5 2017 /Latest
c)	Method of HV Testing :	IS:2071 (Part I, II & III)
d)	H.V. Test Techniques :	IEC-60060
e)	Partial discharge measurements :	IEC-60270
f)	Electromagnetic compatibility (EMC) :	IEC:61000
g)	Radio Interference test on high- : voltage insulators	IEC:60437
h)	Method for testing uniformity of coating on zinc coated articles :	IS:2633
i)	Terminal connectors :	IS:5561
j)	Precision and semi-precision : hexagon bolts, screws, nuts & lock nuts	IS:1364
k)	Insulation Coordination :	IEC 60071-1, 2
l)	Specification Part-4 for Surge : Arrestors without gap for AC system.	IEC 60099-4
m)	Recommended Practice for hot : dip galvanizing of iron and steel.	IS : 2629
n)	Classification of degrees : of protection provided by enclosures of Electrical equipment .	IS/IEC : 60529/2001
o)	Polymer Hollow Insulators :	IEC: 61462

p) Porcelain Hollow Insulators : IEC: 62155

REFERENCE STANDARDS	NAME AND ADDRESS
IEC	International Electro Technical commission Bureau Central de la commission,Electro Technique International,1Rue de verembe, Grneva, Switzerland.
ISO	International Organisations for Standardization Danish Board of Standardization, Danish Standardising sraat, Aurehoegvaj - 12 DK - 2900, Heelprup DENMARK
IS	Indian Standard Bureau of India Standards Manak Bhavan, 9, Bahadur shah Zafar Marg New Delhi - 110 002 INDIA.

NOTE :

i) For the purpose of this specification all technical terms used herein after shall have the meaning as per IEC Specification.

ii) For the parameters of the Arrestor which are not specified in IEC specification for Surge Arrestors , the provisions of IS 3070 (Part-I) /IS 15086 Part-4:2017 shall be applicable.

iii) The equipment/material must be manufactured and tested as per latest IS/IEC/ASTM/ IEEE/CBIP standards, active as on date of approval of drawings/GTP/PO whichever is approved at the latest".

2.2 The electrical installation shall meet requirements of Indian Electricity Rules, 1956 and IS-1886, "Code of practice for installation & maintenance of Transformers" as amended till date.

2.3 **DUTY REQUIREMENTS:**

2.3.1 The surge arresters shall be of heavy duty, Station class and Gap less type without any series or shunt gaps.

2.3.2 The surge arresters shall be adequately designed to withstand specified dynamic over-voltages. The Bidder shall furnish the data /calculations for the duration and over-voltage factors which the offered arresters can withstand after absorption of energy equivalent to specified energy discharge class. (Two long duration discharge current impulses).

2.3.3 The surge arresters shall be capable of discharging over-voltage occurring during switching / tripping off unloaded transformers, reactors and long lines and phase to ground faults.

2.3.4 Surge Arrester housing shall conform to technical requirements in Clause 3.6 of this section.

2.3.5 The reference current of the surge arresters shall be high enough to eliminate the influence of grading and stray capacitance on the measured reference voltage of the arrester units (with designed grading system).

2.3.6 Arresters shall be designed with sufficient cantilever strength to meet stress due to specified wind speed and seismic acceleration.

2.3.7 Arresters shall require no routine maintenance upkeep, or attendance, except as required to remove pollution contaminants.

2.3.8 The Surge arresters are provided to protect the following equipment whose insulation levels are indicated as under:

Equipment to be protected	Lightning impulse (kVp) for 420kV system	Switching Surge (kVp) for 420kV system	Lightning impulse (kVp) for 245kV system	Lightning impulse (kVp) for 145kV system	Lightning impulse (kVp) for 36kV system
1	2	3	4	5	6
Power transformer	± 1300	± 1050	± 950	± 550	±170
Instrument transformer	±1425	± 1050	± 1050	± 650	± 170
CB/Isolator Phase to ground	± 1425	± 1050	±1050	± 650	±170
CB/Isolator Across open contacts	± 1425 (-/+ 240)	± 900 (-/+ 345)	± 1200	±750	± 170

2.3.9 The duty cycle of Circuit Breakers taken in this substation shall be 0-0.3 secs-CO-3 minutes-CO. The Surge Arrester shall be suitable for such circuit breaker duties in the system.

3. CLIMATIC CONDITIONS: Refer Clause-2 of Section-I, General

4.0 PRINCIPAL PARAMETERS:

The Surge Arrestors offered under this specification shall conform to the parameters given below.

Sl No.	Particulars	220 kV	132 kV	33 kV
1.	Nominal System voltage(kV rms)	220	132	33
2.	Highest System voltage(kV rms)	245	145	36
3.	1.2/50 micro second impulse voltage			

	withstand level			
	a. Transformers and Reactors(kVp)	950	550	170
	b. Other equipment and lines(kVp)	1050	650	170

Sl No.	Particulars	220 kV	132 kV	33 kV
4.	Minimum prospective symmetrical fault current for 1 second at Arrester location (kA rms)	50	40	31.5
5.	Anticipated levels of temporary overvoltage and its duration			
	a) Voltage (p.u.)	1.5/1.2	1.5/1.2	1.5/1.2
	b) Duration (Seconds)	1/10	1/10	1/10
6.	System frequency (Hz)	50 ± 2.5	50 ± 2.5	50 ± 2.5
7.	Neutral Grounding	E.E.	E.E	E.E
8.	Number of Phases	Three	Three	Three

Note: E.E. = Effectively Earthed.

5.0 TECHNICAL REQUIREMENTS :

5.1.1 The Surge Arrestors shall conform to the technical requirements as per Annexure -I.

5.1.2 Guaranteed Technical Particulars of the Surge Arrestors offered shall be furnished in Annexure - II.

5.2 The energy handling capability of each rating of Arrester offered , supported by calculations, shall be furnished in the offer.

5.3 The Surge Arrestors shall be fitted with pressure relief devices and arc diverting ports and shall be tested as per the requirements of IEC specification for minimum prospective symmetrical fault current as specified in clause 4.0 (4).

5.4 The Grading ring on each complete Arrestor for proper stress distribution shall be provided if required for attaining all the relevant technical parameters.

5.5 **PROTECTIVE LEVELS:**

The basic insulation levels and switching impulse withstand levels of the lines and equipment to be protected, have been specified in clause 4.0, " Principal Parameters". The protective characteristics of the Arrestors offered shall be clearly specified in the schedule of Guaranteed Technical Particulars.

5.6 GENERAL REQUIREMENTS:

5.6.1 The materials and components not specifically stated in this specification but which are necessary for satisfactory operation of the equipment are deemed to be included in the scope of supply unless specifically excluded.

5.6.2 Unless otherwise brought out separately by the Bidder in the schedule of deviations the Surge Arrestors offered shall conform to the specification scrupulously. All deviations from the specification shall be brought out in the schedule of deviations . The discrepancies between the specification and the catalogues or literature submitted as part of the offer shall not be considered as valid deviations unless specifically brought out in the schedule of deviations.

5.6.3 Any deviation which has not been specifically brought out in the schedule of deviations of the Bid Proposal Sheets , shall not be given effect to. The deviations brought out in the Schedule shall be supported by authentic documents, standards and other references.

5.6.4 Each individual unit of Surge Arrestor shall be hermetically sealed and fully protected against ingress of moisture. The hermetic seal shall be effective for the entire life time of the Arrester and under the service conditions as specified The Bidder shall furnish sectional view of the Arrestor, showing details of sealing employed.

5.6.5 The bidder shall furnish in the offer, a sectional view of pressure relief device employed in the Surge Arrestors offered.

5.6.6 The Surge Arrestors shall be suitable for hot line washing.It has to be ensured that there is no accumulation of humidity in the design.

`5.7 CONSTRUCTION:

5.7.1 All the units of Arrestors of same rating shall be interchangeable without adversely affecting the performance.

5.7.2 The Surge Arrestors shall be suitable for pedestal type mounting.

5.7.3 All the necessary flanges, bolts, nuts, clamps etc., required for assembly of complete Arrestor with accessories and mounting on support structure to be supplied by the Purchaser, shall be included in bidder's scope of supply.

5.7.4 The drilling details for mounting the Arrestor on Purchaser's supporting structures shall be supplied by the Supplier.

5.7.5 The minimum permissible separation between the Surge Arrestor and any earthed object shall be indicated by the Bidder in his offer.

5.7.6 The surge arresters shall be of hermetically sealed units, self-supporting construction suitable for vertical mounting on a galvanized horizontal steel structure /steel tubular plat form.

5.7.7 The non-linear blocks shall be of sintered metal oxide material. These shall be provided in such a way as to obtain robust construction with excellent mechanical electrical properties even after repeated operations.

5.7.8 The surge arresters shall be fitted with pressure relief devices suitable for preventing shattering of porcelain housing in the event of arrester failure

5.7.9 Arresters shall incorporate an anti-contamination feature to prevent arrester failure consequent to uneven voltage gradient across the stack, in event of contamination of the arrester porcelain/Polymer.

5.7.10 A corrosion-resistant nameplate with clearly legible lettering shall be permanently attached to each surge arrester at an easily visible place. The name plate shall be readily legible from the ground and shall meet the requirements of IEC recommendation.

5.7.11 The arrester assembly shall consist of arrester unit, line terminal, earth terminal, insulating Base and other hardware required for installation.

5.7.12 The arrester offered shall be suitable for hot line washing. A copy of the instructions for hot line washing shall be furnished with the bid.

5.7.13 The manufacturer shall submit data on rejection rate of Zno blocks during manufacturing/operation for the past three years.

5.7.14 The nameplate shall conform the requirements of IEC incorporating the year of manufacture.

5.7.15 The heat treatment cycle details along with necessary quality checks used for individual blocks along with insulation layer formed across each block are to be furnished. Metalizing coating thickness for reduced resistance between adjacent discs is to be furnished with additional information schedule of bid proposal sheets along with procedure for checking the same. Details of thermal stability test for uniform distribution of current on individual disc is to be furnished

5.7.16 Arrester housing shall be manufactured and tested in accordance with IEC:62155:2003.

5.7.17 Housing insulators shall be manufactured from high quality **porcelain/polymer.**

Porcelain used shall be homogenous, free from laminations, cavities and other flaws or imperfections that might affect the mechanical or dielectric quality and shall be thoroughly vitrified and impervious to moisture.

Glazing of the porcelain shall be uniform brown in colour, free from blisters, burrs and similar other defects. Housing shall incorporate suitable profile preferably of self cleaning type (aerofoil design) meeting with IS/IEC 60168/2008 and with required

minimum creepage distance in order for arrester to function satisfactorily under specified pollution environment.

5.8 Porcelain Housing :

5.8.1 All porcelain housings shall be free from lamination cavities or other flaws affecting the maximum level of mechanical and electrical strength.

8.2 5.8.2 The porcelain shall be well vitrified and non-porous.

8.3 5.8.3 The creepage distance of the Arrestor housing shall be as per Annexure -I.

5.8.4 The porcelain petticoat shall be preferably of self-cleaning type (Aero foil design). The details of the porcelain housing such as height, angle of inclination, shape of petticoats, gap between the petticoats, diameter (ID and OD) etc. shall be indicated in a detailed drawing.

5.8.5 The Arrestor housing shall conform to the requirements of IS :3070 specification.

5.8.6 Polymer Housing

5.8.6.1 Polymer Arrester design to be based as per IEC 60099-4 of 2014 and IEC 60099-5 Edition 3.

Surge arresters for 220 kv system maybe of tube/Cage design only

Surge arresters for 132 kv system maybe of tube/cage design only

Surge arresters for 33 kv system maybe of tube/cage/wrap design only

Type A: Tube design Arresters: These arresters use hollow core polymer insulator with an intended included gas volume. It must consequently equipped with pressure relief devices and sealing system. The housing is usually a non –ceramic composite housing, formed e.g. from a tube of fiber glass reinforced plastic (FRP) moulded with silicone rubber weather sheds. Tube design is adopted for high cantilever mechanical strength.

Tube design of surge arresters shall be fitted with pressure relief devices and shall be tested as per the requirement of IEC 60099-4 specification for minimum prospective symmetrical fault current of 50kA/40kA/31.5kA rms for 200 mSec. The bidder shall furnish in the offer sectional view of the pressure relief device employed in the surge Arresters offered for Tube design.

Type B2: CAGE design- Void less construction for Surge Arresters without any gas volume included: The stack of MO resistor elements is clamped by FRP rods. The MO resistor elements act as part of the mechanically supporting structure, and the FRP rods form an open cage. The outer sheds should be directly moulded to the modules, using silicone rubber. These arresters do

not have any pressure relief device and are of a soild type with no enclosed volume of gas. If the resistors fail electrically, an arc is established within the arrester. The arc causes heavy evaporation and possibly burning of the housing and/or internal material. These arresters short circuit performance is determined by their ability to control the cracking or tearing-open of the housing due to the arc effects, thereby avoiding violent shattering.

The rain sheds/ petticoats shall be moulded directly on the CAGE construction and should not be pre-moulded push fit/ slip on type. Zinc Oxide blocks assembled in a pre moulded Tube and sealed at ends will not be accepted.

Type B1: Wrap design- Void less construction: The Mechanically supporting part of the housing is formed by a wrapped FRP using Prepreg glass rovings or pre-impregnated bands that are wound around the MO resistor stack and cured. The resulting wrap, surrounding the varistor stack, may be completely closed or to have open "windows". (moulded with silicone rubber for relieving the short circuit.) The wrap may also consist of a prefabricated FRP tube, which is pushed over the stack of MO resistor elements and where in the gap between MO resistors and FRP tube is then filled by some elastic material.

Cantilever force for 245kV and below surge Arresters is not less than 150kg with deflection not exceeding 200mm. Cantilever force for 400kV cage design surge Arresters not less than 350kgf with deflection not exceeding 200mm.Cantilever force for 400kV Tube design surge Arresters not less than 350kgf with deflection not exceeding 200m

5.8.6.2 **Housing of Polymer Surge Arrester**: Silicone rubber used for arrester housing must be resistant to tracking and erosion, and stabilized against UV radiation. Silicon rubber housing shall be free from lamination cavities or other flaws affecting the mechanical and electrical performance. Surge Arrester shall not fail due to housing contamination. The adhesion between the polymeric housing and the metal oxide resistors or any other metallic or non metallic parts inside the housing must be strong enough, homogenous, robust and resistance to thermal cycles and environmental stresses. The polymer sheds shall be preferably of self cleaning type (Aerofoil type) The arrester housing details such as shed profile, angle of inclination, gap between sheds should be as per IEC/TS 60815-3 Edition 1 2008. Insulation levels of Arrester housing shall conform to the requirements of latest IEC 60099-4 and type tests on Housing shall be done including additional tests as per IEC 60099-4 and IEC 61462 -2007 ed2.

5.9 Galvanisation, Nickel Plating etc :

5.9.1 All ferrous parts exposed to atmosphere shall be hot dip galvanised as per IS:2629 as amended from time to time., Tinned copper/ brass lugs shall be used for internal wiring of discharge counter. Screws used for electrical connections shall be either made of brass or nickel plated.

5.9.2 Ground terminal pads and name plate brackets shall be hot dip galvanised.

5.9.3 The material shall be galvanised only after completing all shop operations.

5.9.4 Minimum weight of zinc coating shall be 900 g/sq.m and minimum thickness of zinc coating shall be 127 microns

5.10 Accessories and Fittings :
Surge Counter / Discharge Counter:

5.10.1 Discharge counters shall be provided for 220 kV ,132 kV and 33kV Arrestors. The discharge counter shall be provided with milli-ammeter for measuring the leakage current and shall not require any DC or AC Aux. Supply. It shall be suitable for out door use. The installation of discharge counter shall not adversely affect the Arrestor performance.

5.10.2 The discharge counter shall register operation whenever lightning or any other type of surge strikes the Surge Arrestor.

Self contained discharge counters suitably enclosed for outdoor use. Fully weather proof and sealed for life and requiring no auxiliary or battery supply for operation shall be provided with a suitable cyclo metric counter for each pole of surge arrester to record the number of discharges through the surge arrester. The discharge counters shall be capable of counting at least 9999 discharges. There shall not be provision to reset the counter. A leakage current meter with suitable scale range to measure the leakage current of the surge arrester shall also be provided with the discharge counter. Surge counter comprising discharge counter and leakage current meter shall conform to environment protection corresponding to IP 67.
The material used in the discharge counter shall be such that it can safely withstand discharge current of the surge arrester. The casing of the surge counter shall be aluminium.
Discharge counter shall be so designed as to be mounted on the supporting structure. The connecting wire used between the discharge counter and the surge arrester shall be supplied along with the equipment. The surge monitor shall be mounted at a height of 1.5m above ground level and away from direction of pressure release vent.

5.10.3 All necessary accessories and earthing connection leads between the bottom of the Arrestor and the discharge counter shall be in the bidder's scope of supply. The connecting lead between discharge counter and Surge Arrestor PVC insulated multiple stranded flexible copper wire of size 70 sq.mm and minimum of 3.0 metre length .The discharge counter shall be so designed that the readings of discharge recorded by the counter and the readings of the milli-ammeter shall be clearly visible through an inspection window to a person standing on ground. The structure height shall be such that minimum live point to ground clearance with respect to the voltage shall be maintained. Considering the height of the equipment, proper modular structure arrangements shall be kept in the scope of supplier as per the requirements of APTRANSCO.The minimum height of support provided by the purchaser shall be 2.440m.. The Live Part to Plinth level clearance shall be maintained as follows:

220kv-5500mm
132kv-4600mm

If not maintained above live point clearances with offered type tested Surge Arrestors on structure Height of 2440mm then special stool arrangement shall be supplied along with offered Surge Arrestors by bidder at free of cost.

5.10.4 Terminal connector conforming to IS : 5561 shall be supplied alongwith the arrestor.

5.11 Each Surge Counter shall have terminals of robust construction for connection to earthing lead and these shall be suitably arranged so as to enable the incoming and outgoing connections to be made with minimum bends.

Following performance is desired from the discharge counter:

The discharge counter shall operate for a minimum lightning impulse current (8/20μs) of 100 amps and a maximum of high current impulse 4/10 μs, 100kA peak (2 application).

The discharge counter shall operate perfectly if the interval time of multiple lightning strokes is more than 0.1 second.

The discharge counter shall have negligible effect on the protection level of the surge arrester.

Long duration current impulses corresponding to the discharge class of the surge arrester with which it is to be used.

Normal discharge current corresponding and discharge current of the surge arrester with which it is to be used.

The potential across the discharge counter shall be less than 10V (rms.) under normal conditions.

An inspection window shall also be provided through which the readings can be taken.

A potential free contact shall be provided which shall close whenever the surge is discharged to ground. This contact shall be available for use by APTransco.

The Bidder shall give a detailed description along with block diagrams of the discharge counter.

Functional test reports on discharge counters shall also be submitted along with the tender. The supplier will be responsible for proper functioning of the discharge counters with the surge arresters. The Bidder shall submit along with the tender test certificates of tests conducted to prove that the discharge counter will register every impulse of more than 100 amps and discharge counter with current meter will have practically no effect on the arrester performance and its residual voltage.

5.11.1 The grounding terminals shall be suitable for accommodating Purchaser's grounding connection to 100x16mm steel earth mat. Proper functioning of the Surge Counter with Surge Arrestor shall be ensured by the Supplier.

5.12 Name Plate :

The arrestor shall be provided with non-corrosive legible name plate indelibly marked with the following information.

1. Transmission Corporation of Andhra Pradesh Limited
2. Order no.
3. Manufacturer's name or trade mark and identification no. of the Arrester being supplied.
4. Rated Voltage.
5. Maximum Continuous operating voltage.
6. Type.
7. Rated Frequency.
8. Nominal discharge current.
9. Long Duration discharge class.
10. Pressure relief current in kA rms.
11. B.I.L. of the equipment to be protected.
12. Year of manufacture.

5.13 The Guaranteed Technical Particulars for the material being supplied shall be provided with this bid as specified in the Annexure-II enclosed. The bids received without these particulars will be considered non-responsive.

6.0 TESTS:

6.1 TYPE TESTS :

6.1.1 The Surge Arrestors of the type and design offered should have been fully type tested as per the relevant standards. The date of type tests conducted shall be the latest and not earlier than 10 (Ten) years from the date of bid opening, subject to no alteration in the design. The Bidder shall furnish one set of Type Test reports, along with the offer.

The type tests shall include

S. No.	**Type Tests**
1	Insulation withstand Test on the Arrester Housing. a) Lightning Impulse b) Power frequency c) Switching Impulse (For 400KV)

2	Residual Voltage Test a) Steep Current Impulse b) Lighting Impulse c) Switching Impulse
3	Test to verify long term stability under COV
4	Repetitive charge transfer withstand
5	Heat dissipation behavior verification of test sample
6	Operating duty test
7	Power Frequency voltage Vs.Time Curve
8	Arrester disconnector
9	Short circuit test a) High Current. b) Low Current
10	Bending Test(Cantilever Strength)
11	Seal Leak rate Test
12	Radio interference voltage test & Corona inception & extinction voltage test
13	Test to verify the dielectric withstand test
14	Test of internal grading components
15	IP-67 test on surge Monitor
16	Weather ageing test a. Salt fog test 1000h
b. UV light test 1000h	
17	All applicable test on silicon rubber as per standard IEC-61462and TERT (Tracking & Erosion resistance test) as per ASTM D2303
18	Degree of Protection test on counter/surge monitor
19	Uniformity of Zinc coating for metal parts
20	Seisimic withstand Test

However the purchaser reserves the right to demand repetition of some or all the type tests in the presence of purchaser's representative.

Copies of certificates of the type and special tests conducted (on same model as offered) according to the IS 15086 Part-4:2017 and IEC 61462, IEC 60099-4 standards shall be furnished along with the bid failing which the bid is liable for rejection.

The type & special tests should have been either conducted in a independent institution/laboratory (accredited by NABL).

6.2 Acceptance and Routine Tests :

6.2.1 All acceptance and routine tests as stipulated in the relevant standards shall be carried out by the Supplier in the presence of the Purchaser's representative.

6.2.2 Acceptance tests, whenever possible shall be conducted on the complete arrester unit. The number of samples to be subjected to acceptance tests shall be decided by the Purchaser at the time of actual testing.

6.2.3 The artificial pollution test shall be carried out as per applicable standards. This test shall be considered as a type test.

6.2.4 The special thermal stability test shall be carried out as acceptance test.

6.2.5 **Routine Tests:**

SN	**List of Routine test reports**
1	Measurement of reference voltage
2	Lightning Impulse Residual voltage test
3	Partial discharge test
4	Tests on discharge counter
5	Visual / Dimensional check
6	Leakage check rate test for a duration of 24 hrs to check the water penetration, on any one randomly selected sample from every 50(Fifty) or below nos. of LA offered for inspection, shall be carried out and reports shall be submitted
7	Verticality check on completely assembled surge Arrestor as a sample test ~~on each lot~~

Acceptance tests:

The following tests shall be made on the samples.

S No	List of Acceptance tests
1	Power frequency reference voltage test
2	Lightning impulse residual test on complete arrester /unit of arrester
3	Leakage check test
4	Partial discharge test
5	Visual / Dimensional check
6	Galvanization test on metal parts
7	Functional (operational) tests on surge monitor/counter at nominal discharge currents (a) 100 Amps with 8/20 microsecond wave shape. (b) 10/20 KA with 8/20 microsecond wave shape
8	**Test on insulators** **All routine tests shall be conducted on Polymer housing in accordance to IEC-61462**

Special Acceptance Test:	
9	Special Thermal stability test as per cl.9.2.2 of IEC 60099-4-2009
10	Ageing Test for Zinc oxide blocks as an acceptance test is to carried out on 3 samples for 72 hours at maximum continuous over voltage (MCOV) and a temperature of 115 C. Acceptance norm being Ir (resistive current)/Watt loss shall remain or decrease at the end of 72 Hrs from the value taken after 1 hour of start of test. Watt loss test:
11	Hydrophobicity test
12	Silicone content test (Silicone Content Min 30%)
13	Glass content test for FRP rods and tubes (Glass Content Min 70%)
14	IP 67 on surge monitor
15	Clamps & connectors a) Chemical Test b) Contact resistance test
16	Cantilever test on complete arrester as per requirement of Annnexure-1
17	Adhesive test between Polymer and tube/cage with Ultrasonic sound equipment.

The acceptance tests shall include the galvanisation test on metal parts

Important Note:

1. The tests shall be performed by the Clamps and connectors manufacturer as part of routine tests and the test certificates shall be produced at the time of acceptance tests. The clamps and connectors shall be procured from APTRANSCO approved vendors only.

2. The Manufacturers shall Submit The Test Certificates of Polymer Material issued by Manufacturers (Sub Vendors)during Final Inspection of Lightning Arrestors.(in case of Polymer LAs)

3. The Successful Bidder shall use the Polymer Material from PGCIL Approved Sub Vendors and shall get prior Sub Vendor Approval from APTRANSCO.

4. In case of any Dispute or Discrepancy at a later Date APTRANSCO has the Right to Test materials at Third Party NABL Acredited Laboratory for initiating appropriate Action.

6.2.6 The functional (operational) acceptance tests shall be carried out on the Surge counter by way of checking its operation at the following nominal discharge currents:

a) 100 Amps with 8/20 microsecond wave shape.
b) 10 kA with 8/20 microsecond wave shape.

Surge monitors shall be routinely tested for water dip test at 1.5m for 30 minutes. No water vapors shall be visible on the monitor glass.

7.0 INSPECTION:

i)The Purchaser shall have access at all reasonable times to the works and all other places of manufacture, where the Surge Arrestors are being manufactured and the Supplier

shall provide all facilities for unrestricted inspection of the supplier's works raw materials, manufacture of the equipment, all the accessories and for conducting necessary tests as detailed herein.

ii)The supplier shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of equipment in its various stages so that arrangements could be made for inspection.

iii) The Supplier shall give 15 days (for local supplies) 30 days (for foreign supplies) advance intimation to enable the purchaser to depute his representative for witnessing acceptance and routine tests. All charges in connection with inspection by purchaser's personnel such as travel, accommodation and incidentals shall be borne by the purchaser.

iv)No material shall be despatched from its point of manufacture unless the material has been satisfactorily inspected ,tested and cleared for despatch.

v)The acceptance of any quantity of equipment shall in no way relieve the Supplier of the responsibility for meeting all the requirements of this specification and shall not prevent subsequent rejection if such equipment are later found to be defective.

8.0 QUALITY ASSURANCE PLAN: Refer Clause-3 of Section-I, General

9.0 DOCUMENTATION:

9.1 All drawings shall conform to International Standards Organization (ISO) 'A' series of drawing sheet/Indian Standards Specification. All drawings shall be in ink and suitable for micro filming. All dimensions and data shall be in S.I. Units only.

9.2 List of Drawings and Documents.

The Bidder shall furnish four sets of following drawings along with his offer:

i) General outline drawings of the complete Arrestors with technical parameters.

ii) Drawing showing clearances from grounded and other live objects and between adjacent poles of Surge Arrestors, required at various heights of Surge Arrestors.

iii) Drawings showing details of pressure relief devices.

iv) Detailed Drawing of discharge counters along with the wiring and schematic drawing of discharge counter and meter.

v) Outline drawing of insulating base.

vi) Details of grading ring if used.

vii) Mounting Details of Surge Arrestors with recommended minimum clearances

viii) Details of line terminal and ground terminals.

ix) volt-time characteristics of Surge Arrestors.

x) Details of Galvanizing being provided on different ferrous parts.

xi) The detailed dimensional drawing of porcelain housing such as ID,OD, thickness and insulator details such as height, profile of petticoats, angle of inclination and gap between successive petticoats, total creepage distance etc.,

xii) Polymer Insulator (only for Design 'A' TUBE design of Arresters)

9.2.1. The supplier should within two weeks of placement of order, submit four sets of final versions of all the above said drawings for purchaser's approval. The purchaser shall communicate his approval/comments on the drawings within two weeks from the date of receipt of drawings. The supplier shall if necessary, modify the drawings and resubmit four copies of the modified drawings for purchaser's approval within two weeks from the date of purchaser's comments. The purchaser shall communicate his approval within two weeks from the date of receipt of the revised drawings. After receipt of purchaser's approval, the supplier shall within three weeks, submit 20 prints and two good quality reproducible of the approved drawings for purchaser's use.

9.2.2 The approval of the drawings by the purchaser shall not relieve the supplier for his responsibility and liability for ensuring correctness and correct interpretation of the latest revision of applicable standards, rules and codes of practices.

9.3 TEST REPORTS :

i) Four copies of acceptance test reports shall be furnished to the Purchaser. One copy will be returned, duly certified by the Purchaser and only thereafter shall the materials be despatched.

ii) All records of routine test reports shall be maintained by the Supplier at his works for periodic inspection by the Purchaser.

iii) All test reports of tests conducted during manufacture shall be maintained by the Supplier. These shall be produced for verification as and when requested for by the Purchaser.

10.0 **PACKING AND FORWARDING: Refer Clause-4 of Section-I, General**

11.0 QUANTITY AND DELIVERY REQUIREMENTS : Refer Clause-5 of Section-I, General

12.0 Terminal Connectors:

12.1 CONSTRUCTION DETAILS

12.1.1 Aluminum or aluminum alloy used in the manufacture of terminal connectors shall meet requirements of standards specified herein. Steel bolts and nuts shall be of reputed make.

12.1.2 Terminal connector shall carry full rated current continuously and its temperature shall not exceed eighty five (85) degrees centigrade.

12.1.3 The connector shall have six (6) bolts with nuts to hold the Moose ACSR conductor (31.77mm dia) up to 220kv and for 400kv Twin Moose conductor, with check nuts and

washers. It shall have rating corresponding to those of the equipment or higher and shall be designed most liberally to comply in all respects with temperature rise, resistance, tensile strength and short circuit current withstand capacity tests as specified in IS-

12.1.4 Connector shall have sufficient mechanical strength and shall completely enclose the conductor sand terminal very tightly so that the mechanical tresses set up due to vibration, wind and short circuit current can be withstood. The conductivity of connector shall be high.

12.1.5 All castings shall be free from blow holes, surface blisters, cracks and cavities. All sharp edges and corners shall be deburred and rounded off.

12.1.6 Size of terminal/conductor for which the connector is suitable and the rated current under site condition shall be punched on the connector body.

12.1.7 All current carrying parts shall be designed and manufactured to offer the minimum contact resistance.

12.1.8 Bolts shall be located as close to the conductor as possible and shall be opposite each other. Bolt diameter shall be such as to give the desired clamping and contact pressure.

12.1.9 check nut shall be provided over nut on each bolt to void loosening of nuts due to vibration.

12.1.10 Steel bolts, nuts, washers and check nuts shall be made of appropriate strength.

12.1.11 The drawings of clamps and connectors shall be submitted along with equipment drawings by the successful Bidder for approval.

13 SUPERVISION SERVICES : Refer Clause-6 of Section-I, General

14.0 GUARANTEED TECHNICAL PARTICULARS:

The Guaranteed Technical particulars for the equipment being supplied shall be provided with the Bid as specified in this technical specification at Annexure-I**.** The Bids received without guaranteed technical particulars shall be treated as Non-responsive.

ANNEXURE - I (Surge Arrestors)

Technical Requirements for Metal Oxide (Gapless) Surge Arrestors

No	**Particulars**	**400kV system**	**220kV system**	**132kV system**	**33kV system**
1	Rated voltage of arrester	360	216	120	30

2	Maximum Continuous operating voltage	294 kVrms	184 kVrms	106kVrms	25.5kVrms
3	Installation	Outdoor			
4	Type	Station class,gapless type			
5	No. of columns	Single column			
6	Type of mounting	Pedestal			
7	Connection	Phase to Earth			
8	Nominal discharge current KAp (8/20 Micro seconds)	20kA	10kA	10kA	10kA
9	Arrester designation	SH	SM	SM	SL
10	Long Duration Discharge Class	4	3	3	2
11	Energy dissipation capability (kJ/kV of rated voltage of arrester) Minimum	12 kJ/kV	7kJ/kV	7kJ/kV	4.0kJ/kV
12	Repetitive charge transfer rating Q_{rs} in coulombs	2.4C	1.6C	1.6C	1C
13	Pressure relief class/short circuit current	63kA	50kA	40kA	31.5kA
14	Prospective symmetrical fault current for pressure relief test kA(rms)	63kA	50kA	40kA	31.5kA
15	High current impulse test value (4/10µs) kA_p	100	100	100	100
16	Max Residual voltage (kVp) for				
	a. At Lightning impulse current of 8/20µs kvp				
	At 5kA	-	567	333	85
	At 10kA	860	602	355	90
	At 20kA	915	668	390	104
	At 40KA	1000	-	-	-
	b) Switching impulse current (30-60/100-200 microsec) of KVp				
	At 500 A	700 KVp	-	-	-
	At 1kA	-	496 kVPeak	294 kV_{Peak}	71kVpeak
	At 2KA	730 KVp	-	-	
	c. Steep current impulse(1/20 microseconds) at discharge current in KVp				
	10KA Peak	-	654kVp	386kVp	101kVp

	20KA Peak	1000	-	-	
17	Maximum PD level at 1.05 COV	Less than 10pC			
18	Seismic acceleration	0.3 g horizontal			
19	Maximum RIV at 1.1 times phase voltage (micro V)	500 at 266kvrms	500 at 156KVRMS	500 at 92KVRMS	NA
20	Insulation level of surge Arrester housing				
	a) Power frequency withstand voltage	630 KV RMS	460 KV RMS	275kv RMS	70kv RMS
	b) Lightning impulse withstand voltage	±1425 Kvp	±1050 Kvp	±650 Kvp	±170 Kvp
	c) Switching impulse withstand voltage	±1050 Kvp	NA	NA	NA
	Minimum total creepage distance	13020mm (31mm/kv)	7595 mm (31mm/kv)	4495mm (31mm/kv)	1116mm (31mm/kv)
21	Temporary over voltage withstand in kVp				
	i) (0.1 sec)	432 kV RMS	397kVp	220kVp	48kVp
	ii) (1.0 sec)	414 KV RMS	382kVP	212kVP	46kVp
	iii) (10 sec)	396 KV RMS	367kVP	203kVP	44kV p
22	Terminal connector suitable for ACSR conductor size		Twin moose		Single Moose
23	Take off	Universal			
24	Cantilever Strength (kg) as DLL	350	150	150	150
25	Max Deflection at above cantilever load (mm)	200	200	200	---
26	Whether insulating base and discharge counter with milli ammeter are required	Yes	YES	YES	NA
27	Silicone content	Min 30 %	Min 30 %	Min 30 %	Min 30 %
28	Glass content in FRP rods and tubes	Min 70 % of boron free ECR glass	Min 70 % of boron free ECR glass	Min 70 % of boron free ECR glass	Min 70 % of boron free ECR glass

ANNEXURE II (SURGE ARRESTORS)
GUARANTEED TECHNICAL PARTICULARS

	220kV	132kV	33kV
1 .Name of Manufacturer			

2. Arrestor class & type

3. Applicable Standard

4. Rated Arrestor Voltage(kV)

5. Maximum Continuous
   Operating voltage (MCOV)-(kV)

6. Nominal Discharge Current (kA)
   with 8/20 Micro-second wave

7. Long Duration Discharge

8. Minimum Energy Discharge
   Capability (kJ/kV)

9. Maximum switching current
   impulse residual voltage at
   i) 1000 Amps
   ii) 250 Amps

10. Maximum residual voltage
    with 1 micro second current
    wave at 10 kA (kVp)

11. Maximum residual voltage
    with 8/20 (kVp)
    Micro-second wave

    i) 5 kA
    ii) 10 kA
    iii) 20 kA

12. Safe Fault Current (kA)

13. Lightning impulse withstand
    voltage of Arrestor housing
    with 1.2/50 micro-second wave (kVp).

14. One minute power frequency
    withstand voltage of housing
    (dry/wet) - kv (r.m.s.)

15. High Current short duration impulse
    withstand level with 4/10 micro
    second wave (kA) Peak

16. Pressure Relief Class

17. Over-voltage withstand Capability kV

a) 10 seconds
b) 1 second
c) 0.1 second

18. a) Reference Voltage (kV)
    b) Reference Current (kA)

19. Number of Units per phase
    and rating of each unit

20. Minimum total creepage
    distance (mm)
21. Leakage Current (mA)

22. Total weight of Arrestor (kg)

23. Maximum cantilever strength
    of Surge Arrester (Including Wind Load)

24. Overall Height of Surge Arrestors (mm)

25. Maximum distance recommended
    from equipment to be
    protected by Surge Arrester (mm)

26. Minimum distance between
    grounded object(mm)

27. Minimum distance between
    Arrester phase legs (mm)

28. Any other particulars

Name of firm----------
Signature of bidder----------